Multi-level Coding and Multi-stage Decoding in MAC, Broadcast, and Relay Channel

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***Abstract*— It is known that multi-level coding (MLC) with multi-stage decoding (MSD) approaches the point-to-point constellation constrained channel capacity subject to appropriate choice of code rates at each level. MLC/MSD in the context of multi-node networks has not been previously investigated, and is the subject of this paper. It is shown that MLC/MSD can approach the MAC capacity. Issues related to achieving non-corner points of the capacity region are discussed. A MLC-based superposition scheme for the degraded broadcast channel is discussed and an achievable region outlined. The half-duplex orthogonal degraded relay channel is studied under MLC/MSD. The achievable rate is calculated under two conditions: First we consider end-to-end layer-by-layer decoding, where each layer is decoded at the relay and destination only with help from previous layers. Then, we remove this constraint via a parity-forwarding approach at the relay, where the destination performs two multi-stage decoding operations: one corresponding to the relay-destination link and then one corresponding to the source-destination link. The achievable rate under parity forwarding is shown to be superior.**

I. Introduction

Multilevel coding (MLC) was proposed by Imai and Hirakawa [1], who showed that subject to appropriate choice of code rate for the levels, it can approach the constellation-constrained channel capacity for a point-to-point channel, and therefore with appropriate choice of constellation can approach channel capacity. MLC/MSD is different from the more-popular Trellis Coded Modulation (TCM) [2] in that each address bit b_i is protected by an independent code C^i. At the receiver side, decoding by multistage decoding (MSD) is used where each code C^i is decoded individually, subject to knowing previously decoded levels. Wachsmann et al. [3] present a comprehensive overview of MLC/MSD, including an overview of design rules for the rates of the individual encoders. In a similar manner, the application of MLC/MSD for MIMO channels has been explored [4].

An attraction of multi-level coding is that by decoupling the coding from the dimensionality of modulation, it allows binary codes, whose behavior and design are studied in far more detail than non-binary codes, to be brought to bear on the problem. This attraction applies not just to the point-to-point channel, but also to multi-node networks, producing a compelling motivation to study multilevel codes for network applications, which to the best of our knowledge has been absent in the literature. For certain simple networks, such as the multiple access channel, the degraded broadcast channel, and the degraded relay channel, some of the known encoding and decoding strategies produce new and interesting scenarios with multi-level coding. Examples include superposition coding for the broadcast channel, successive decoding for MAC, and the parity forwarding strategy for the relay channel.

The contributions of this paper are as follows: It is shown that MLC/MSD can approach the capacity of the MAC, and issues related to achieving non-corner points of the capacity region are discussed. A MLC-based superposition scheme for the broadcast channel is discussed and an achievable region outlined. It is shown that for the degraded broadcast channel, MLC/MSD achievable region approaches the capacity region. The half-duplex orthogonal degraded relay channel is studied under MLC/MSD. The achievable rate when each destination stage uses exactly one layer of source signal and one layer of relay signal is calculated. It is shown that when this constraint is removed, MSD can approach the degraded relay capacity.

II. Background and Review

MLC scheme is shown in Fig. 1 where the data is partitioned into m streams. The streams are indexed by $i \in \{1, 2, \ldots, m\}$. Each of the streams is individually and independently encoded to produce a coded stream b_i. Then, at each time instance, a bit from each coded stream is collected to produce the vector $[b_1 \ \ldots b_m]$ that is the label of the constellation point to be transmitted [3].

This constellation point is sent through additive white Gaussian noise (AWGN) channel $P(Y|X)$. Since the modulation mapping is bijective, by data processing identity we have:

$$I(X;Y) = I(b_1, b_2, \ldots, b_m; Y) = \sum_{i=1}^{m} I(b_i; Y|b^{i-1}) \quad (1)$$

where we denote partial vectors $b^{i-1} = [b_1, b_2, \ldots, b_{i-1}]$ in a manner similar to [5]. This equation justifies MLC/MSD. MSD

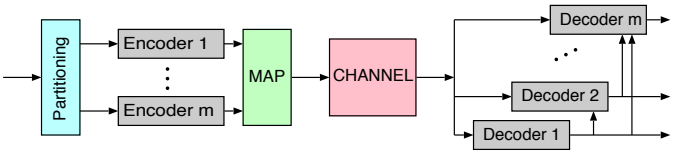


Fig. 1. MLC and MSD in point to point channel.

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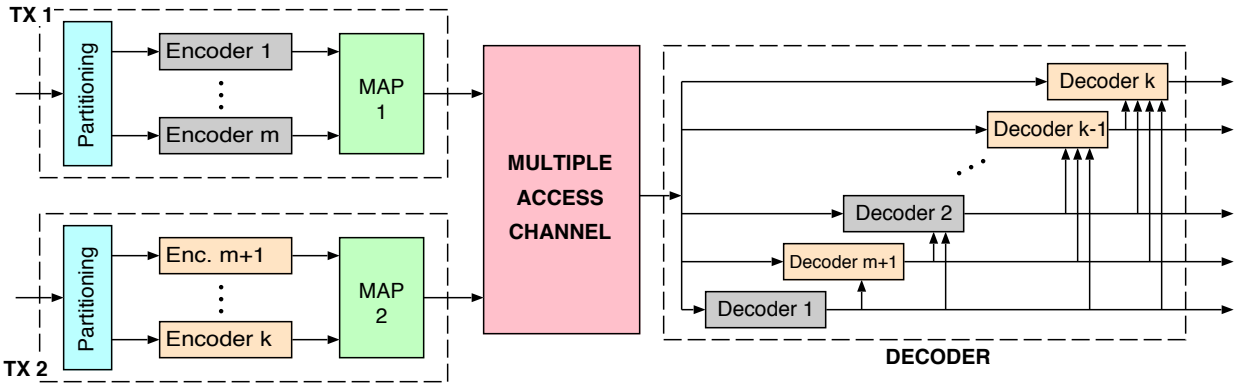


Fig. 2. MLC and MSD in multiple access channel.

is implemented by decoding b_i conditioned on $b_{i-1}, \ldots, b_1$. Subject to choosing the appropriate rates shown by the mutual information in (1), the constellation constrained capacity can be achieved, which itself, subject to appropriate choice of constellation according to SNR, can approach the channel capacity.

Remark 1: Consider an equivalent channel between the individual b_i and the output Y. Since the inputs to b_i have been encoded independently, this multi-input single-output channel is nothing but a MAC. Multi-stage decoding is nothing but decoding at the corner points of this equivalent MAC, thus the rate restrictions for multi-stage decoding can be interpreted in terms of the corner points of a MAC. This insight also shows that multi-level coding can achieve the constellation constrained capacity of the point-to-point channel even outside of the rate constraints shown by (1), but then it will require joint decoding of the streams. This equivalent channel idea will prove useful in the next section.

III. The Multiple Access Channel

The capacity region of the MAC is the set of all achievable rates R_1 and R_2 such that

$$R_1 \leq I(X_1; Y|X_2) \tag{2}$$

$$R_2 \leq I(X_2; Y|X_1) \tag{3}$$

$$R_1 + R_2 \leq I(X_1, X_2; Y) \tag{4}$$

The corner points of the capacity region can be achieved by decoding one of the transmitter's message first and then decoding the other transmitter message taking into account the decoded message of the first transmitter decoded message. The other corner point is achieved by reversing the order of the decoding and any point on the line connecting these two corner points can be achieved by either joint decoding or by time sharing.

Having a closer look at the third bound in the capacity region for the sum rate and using the chain rule we can write the following

$$R_1 + R_2 \leq I(X_1, X_2; Y) = \sum_{i=1}^{K} I(b_i; Y|b^{i-1}) \tag{5}$$

where X_1 and X_2 are the constellation points chosen from the Mapper of the first and second transmitters based on the address bits $[b_1, b_2, \ldots, b_m]$ and $[b_{m+1}, b_{m+2}, \ldots, b_k]$ respectively. Equation (5) produces one of the corner points of the MAC capacity region. Decoding b_i in the opposite order produces the other corner point. Each of these corner points requires a particular rate assignment between the encoders. Therefore we have the following result:

Theorem 1: Multi-level encoding and multi-stage decoding can achieve the constellation constrained capacity of the Gaussian MAC.

Remark 2: Since the encoders for the different levels are independent, the operation of this system is equivalent to a K-user MAC whose inputs are the constellation labels of the two users. It is interesting to note that the rate design for our 2-user MAC involves a corner point of this K-user MAC, where the bits of one user are decoded, followed by the bits of the other user. However, the equivalent MAC shows us that other rate designs are also possible that meet the capacity outer bound, involving any ordering of the decoding of the bits of the two users. However, any mixed ordering may or may not meet the corner points of the 2-user MAC (it may produce another point on the sum-rate boundary).

With the decoding order $b_1 \rightarrow \ldots \rightarrow b_K$, the rate of encoder i is $I(b_i; Y|b^{i-1})$. This set of rates corresponds to a corner point of the rate region of the K-user MAC. It is easy to see, however, that this particular decoding order is not required, and in fact $K!$ decoding orders are possible, each corresponding to a corner point of the capacity region of the K-user MAC. This provides some level of flexibility in the rate assignment to encoders. In addition, the various decoding orders that allow intermixing of the order of the first and second user (in a 2-user MAC, as shown in Fig. 2 with a certain decoding order) will result in achieving multiple points on the boundary of the rate region via successive decoding, many of them not on

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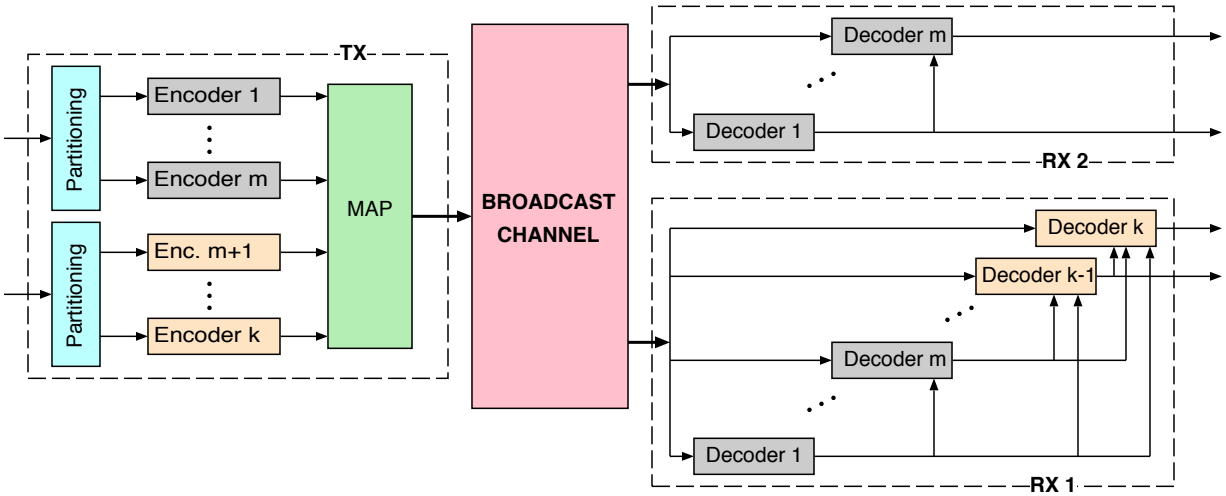


Fig. 3. MLC and MSD in broadcast channel.

the corner points of the 2-user MAC region. Thus, multi-level coding can allow successive decoding to achieve non-corner points of the MAC capacity region.

Remark 3: MLC/MSD was shown to achieve the constellation-constrained capacity. An appropriate choice of the constellation, with respect to the channel SNRs, can then allow the system to approach the Gaussian MAC capacity. This can be formalized into a theorem via a sequence of successively denser constellations.

IV. The Broadcast Channel

For the degraded broadcast channel, superposition coding and successive decoding is known to achieve capacity [6] and the corresponding rate constraints are given by:

$$R_1 \leq I(X;Y_1|U) \tag{6}$$

$$R_2 \leq I(U;Y_2) \tag{7}$$

where U is an auxiliary random variable representing the bit streams of the weak receiver $[b_1,\ldots,b_m]$ and X represents the transmitted symbol based on all the address bits $[b_1,\ldots,b_K]$.

Multilevel coding and multi-stage decoding for the broadcast channel is depicted in Fig. 3, where the data for the first and second receivers are partitioned into K streams and encoded independently. The data stream of the first receiver (the weak receiver) are partitioned into streams $1,\ldots,m$ and the data stream of the second receiver (the strong receiver) is partitioned into the streams $m+1,\ldots,K$. Depicting the auxiliary random variable $U=[b_1,\ldots,b_m]$, we have: $U \rightarrow X \rightarrow Y_1 \rightarrow Y_2$. Furthermore, because the mapping (modulation) is bijective, we can use the data processing inequality to show that in the mutual information expressions, X can be replaced with $[b_1,\ldots,b_K]$.

Applying the chain rule to the right hand side of the three bounds in the capacity region:

$$I(X;Y_1|U) = \sum_{i=m+1}^{K} I(b_i;Y_1|b^{i-1}) \tag{8}$$

$$I(U;Y_2) = \sum_{i=1}^{m} I(b_i;Y_2|b^{i-1}) \tag{9}$$

where the chain rule can be applied in any order. The previous equations justify MLC/MSD to approach the constellation-constrained capacity of the broadcast channel. The weak receiver decodes it's first m streams and the strong receiver decodes the first m streams and then decodes the rest of the K streams.

The rate design of the individual encoders should be less than the mutual information of the corresponding channels. Thus we have the following result:

Theorem 2: For the degraded broadcast channel, multi-level encoding and multi-stage decoding can achieve the following rates:

$$R_2 \leq \sum_{i=1}^{m} I(b_i;Y_2|b^{i-1}) \tag{10}$$

$$R_1 \leq \sum_{i=m+1}^{K} I(b_i;Y_1|b^{i-1}) \tag{11}$$

To achieve this rate, the individual encoders must operate at rates $I(b_i;Y_2|b^{i-1})$ for $i=1,\ldots,m$ and at rates $I(b_i;Y_1|b^{i-1})$ for $i=m+1,\ldots,K$.

For the degraded broadcast channel, it has not been possible to establish that MLC/MSD can achieve the entire capacity region. Part of the reason is that the equivalent channel from the multiple inputs of the mapping (modulation) to the two outputs of the channel constitute an equivalent interference channel, whose capacity at the present is not known.

At the moment, for the broadcast channel the question of approaching the modulation-limited capacity via multi-level

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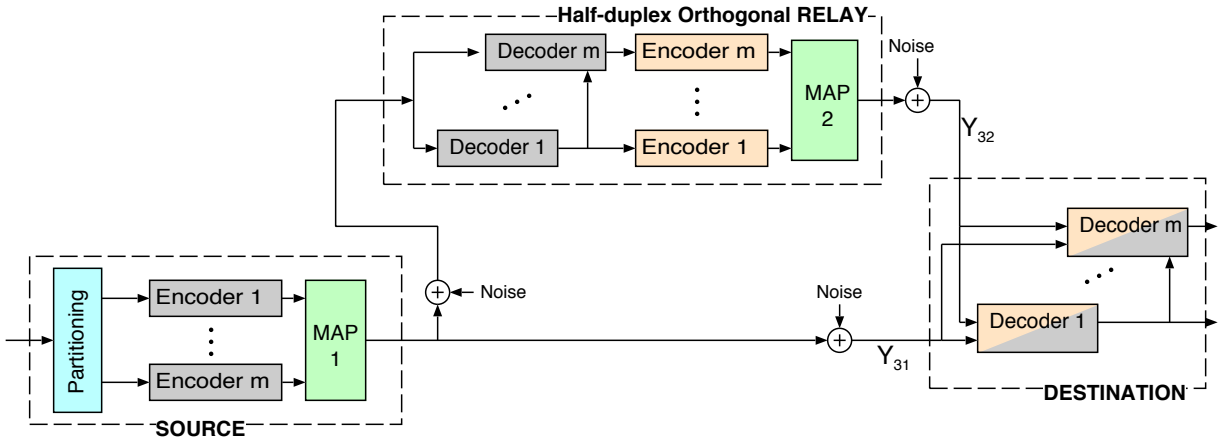


Fig. 4. MLC and MSD in the half-duplex orthogonal relay channel

coding remains open. But it is possible to extend the simple region of Theorem 2 by noticing that given a mapping with K levels, the choice of m as well as the ordering of the multi-stage decoding at each receiver remains open. Therefore we have the following result:

Theorem 3: An achievable region for the degraded broadcast channel with MLC/MSD consists of the convex hull of all rate pairs shown in Equations (10) and (11), subject all choices of $m = 0, \ldots, K$ and for each m subject to all permutation of encoding/decoding orders of the bits.

Remark 4: In order to simplify the presentation, Fig. 3 is presented in a slightly simplified form. In general, it is not necessary that the total K levels be divided between the two users such that each user has an integer number of levels. It is also possible that one of the levels has data partly for one user and partly for the other. In this case, the decoder for that layer must be operable at both users, thus it must be decodable at the weaker user. This is equivalent to "giving" some of the rate of the weak user to the strong user, and it is also equivalent to achieving non-vertex points in the achievable region mentioned above.

V. The Relay Channel

For simplicity, we consider the half-duplex orthogonal relay channel, consisting of two (orthogonal) time slots per codeword. In the first time slot, the source transmits X_1, resulting in the reception of Y_2 at the relay and Y_{31} at the destination. It is assumed that the relay channel is degraded, and a decode-forward strategy is used, therefore the relay is able to decode, re-encode, and transmit a signal X_2 in the second time interval, resulting in the reception of Y_{32} at the destination. The destination then decodes (Y_{31}, Y_{32}) to arrive at the transmitted data.

The achievable rate for this relaying strategy is the minimum of two mutual informations involving decoding at the relay and the destination. The relay decoding condition is:

$$R_{12} \leq I(X_1; Y_2) \tag{12}$$

Applying the chain rule and the mapping between the address bits $[b_1, b_2, \ldots, b_m]$ and the transmitted symbol X_1

$$R_{12} \leq \sum_{i=1}^{m} I(b_i; Y_2 | b^{i-1}) \tag{13}$$

We now consider two relay multi-level codes and the corresponding destination multi-stage decoding.

A. End-to-End Layered Decoding

In this strategy, we require the destination to decode one layer of relay and source codes at a time. In other words, this strategy requires that each layer of the relay code corresponds exactly to one layer of the code transmitted at the source. A by-product of this condition is that the number of layers are the same at the source and the relay, and also each layer of the destination decoding is driven by exactly one layer of code from the source, and one layer of code from the relay (see Fig. 4).

Under this scenario, the MAC cutset bound for layer i can be evaluated with the chain rule as follows:

$$r_i < I(b_i; Y_{31} | b^{i-1}) + I(\hat{b}_i; Y_{32} | \hat{b}^{i-1}) \tag{14}$$

where r_i is the rate allocated to layer i, and $\hat{b}_i$ are the random variables representing the inputs to the mapping at the transmit port of the relay. Note that we specify the cut-set bound for each layer separately because each of the layers have to be individually decoded both at the relay and the destination without the help of "future" layers. Thus, overall we have:

Theorem 4: The rate achievable by the end-to-end layered decoding strategy for the half-duplex orthogonal relay channel is:

$$R \leq \sum_{i=1}^{m} \min \left\{ I(b_i; Y_2 | b^{i-1}), I(b_i; Y_{31} | b^{i-1}) + I(\hat{b}_i; Y_{32} | \hat{b}^{i-1}) \right\} \tag{15}$$

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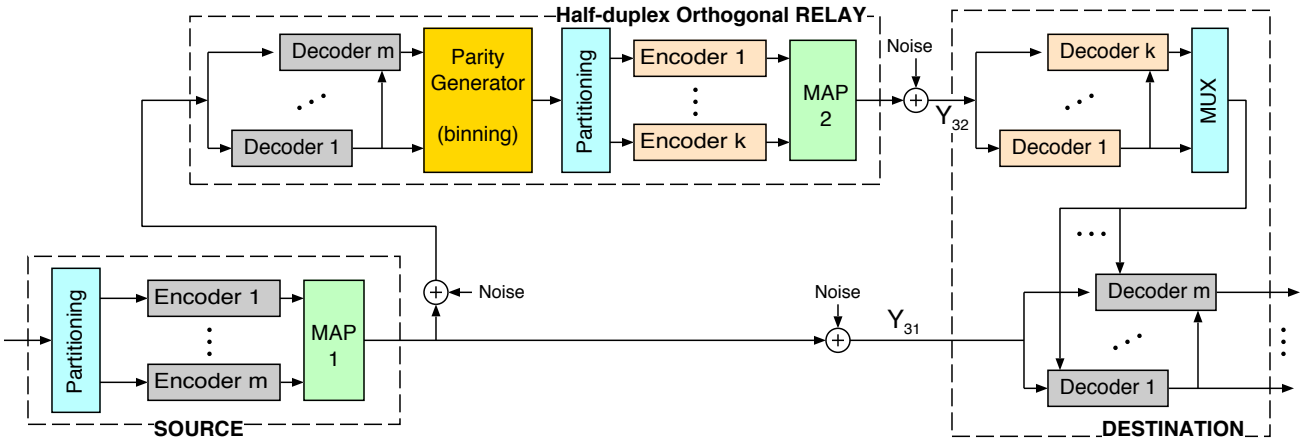


Fig. 5. MLC and MSD in the half-duplex orthogonal relay channel with non-equal levels

This method has the condition that the encoders at the source and relay have essentially the same structure, and the end-to-end system can be simply layered and decoded layer-by-layer. However, in general this constraint could reduce the capacity.

B. Two-stage Decoding at Destination

In this method, we allow the source-relay and relay-destination links to operate with different multi-layer codes, and no longer require that each layer can communicate end-to-end only dependent on previous layers (see Fig. 5). To achieve this end, we use a parity-forwarding strategy at the relay which is inspired by the binning strategy of compress-forward. This strategy can be shown to achieve the decode-forward capacity in degraded relay channels [7].

The idea of parity forwarding is that the relay-destination link allows reliable communication at rate $I(X_2; Y_{32})$, thus we can send that many additional parity bits to the destination. Each layer at destination can pick up a subset of these parity bits, which will add to the total mutual information available at that layer at the decoder, thus allowing the decoding to be completed. Under this strategy, we have the following result:

Theorem 5: The following rate is achievable for the half-duplex orthogonal decode-forward relay, allowing two-stage decoding at the destination:

$$R \leq \min \Bigg\{ \sum_{i=1}^{m} I(b_i; Y_2 | b^{i-1}), \qquad (16)$$
$$\sum_{i=1}^{m} I(b_i; Y_{31} | b^{i-1}) + \sum_{j=1}^{K} I(\hat{b}_j; Y_{32} | \hat{b}^{j-1}) \Bigg\}$$

This achievable rate is higher than the previous case, because the design rates of the individual encoders at each layer are less constrained. The additional mutual information provided to each decoding layer at the destination is exactly as much as needed without additional constraints arising from the structure of the relay-destination link.

VI. Conclusion and Future Work

This paper studied MLC/MSD for MAC, broadcast, and relay channels, calculating achievable rates, and showing that for the MAC channel the constellation constrained capacity can be met with MLC/MSD. In the case of degraded broadcast channel, an achievable rate region was found, however, the question of meeting the degraded capacity outer bound remains open. For the half-duplex orthogonal degraded relay channel under decode-forward, two achievable rates were obtained with and without the layered end-to-end decoding constraint.

Ongoing work involves superposition strategies for non-degraded broadcast channels, and also non-orthogonal half-duplex as well as full-duplex relay channels. Layer-by-layer capacity curves for simple examples in a manner similar to [1], [3] are being simulated.

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